MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED: DELAY XM92 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, XM92, Assembly, loaded with Cartridge, Delay, M155, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS28741	- Hose Assembly, Detachable End Fitting Medium Pressure

DRAWINGS

Munitions Command

F10551995	- Initiator, Cartridge Actuated, Delay, XM92, Assembly
C8798142	- Box, Packing, Ammunition for Initiator, Delay, Cartridge Actuated, M10, M30A1, M31, M32A1, M42 M43A1, M72, XM92 or M114 with Cartridge in Fiber Container, M398
C8798143	- Container, Ammunition, Fiber, M398 for Initiator, Delay, M31, M32A1, M42, M43A1, M72, XM92 or M114 with Cartridge
IEL-10551995	- Inspection Equipment List for Initiator, Cartridge Actuated, Delay, XM92, Assembly.

FSC 1377

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2.1 (Continued)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing Pin Release.- The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds, has withdrawn the initiator pin 3/4 - inch, maximum.

3.3 Initiator Pin Security.- The initiator pin of the assembled initiator, less cartridge, cartridge retainer and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin.

3.4 Breakaway Torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch - pounds, without loosening or deformation.

3.5 Residual Magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size no. 4 in accordance with MS28741.

TABLE I

Temperature	Pressure (PSI)	Ignition Delay (See 6.3)
-65° to -70°F	500 to 1500	.650 Sec (Max)
65° to 75°F	600 to 1500	----------
200° to 205°F	700 to 1700	.375 Sec (Min)

3.6.2 The initiator shall not fail to fire.

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3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship.- The requirements for Workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal Defects.- The component shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign Matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection Provisions.-

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of Lot.- A lot shall consist of an identificable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with acceptable cartridges.

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4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. All non-conforming material shall be rejected. The total number of allowable defects per class for either major or minor classification shall not exceed the acceptable number of Table II. Four power magnification maybe used for inspection of such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% examination	-	-	-	-
301-800	50	0	1	1	2
801-1800	90	1	2	2	3

4.2.2.1 Cartridge, Delay, M155, Dwg D10521632, covering a detail of Drawing F10551995.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defects	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign Matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Initiator, Cartridge Actuated, Delay, XM92, Dwg. F10551995

CRITICAL	
1. Any part missing	Xray1/& Visual
2. Improper assembly	Xray1/& Visual
3. Marking missing, incorrect or illegible	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.2.2.2 (Continued)

Defect	Method of Inspection
MAJOR	
101. Length of initiator pin extension	Gage
102. Length from face of initiator cap to end of gas port, max	Gage
103. Torque application of cartridge retainer with firing pin housing, improper	Gage
104. Torque application of initiator chamber with cap, improper	Gage
105. Foreign Matter	Visual
MINOR	
201 Total length from end of cap to end of gas port, max	Gage
202 Total length	Gage
203 Height of boss, max	Gage
204 Location of label incorrect	Visual
205 Set screw missing or loose	Visual
206 Coating compound missing from set screw	Visual

4.2.2.3 Fiber Container, Dwg. C8798143

CRITICAL

None defined

MAJOR

None defined

MINOR

201 Gap between body and cover, max	Manual
202 Contents loose	Manual
203 Metal ends loose or distorted	Manual
204 Cut or damaged through all impregnated layers	Visual
205 Marking missing, incorrect or illegible	Visual
206 Tape incomplete or badly wrinkled	Visual
207 Length of tear tab, min	Visual
208 Color of tape incorrect	Visual

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4.2.2.4 Box, Packing, Dwg. C8798143

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201	Contents loose	Manual
202	Hardware missing, loose, broken or improperly engaged.	Manual & Visual
203	Strapping missing, loose, broken or improperly assembled	Manual & Visual
204	Contents of box exposed	Visual
205	Box damaged	
206	Marking missing, incorrect or illegible	Visual
207	Number of units incorrect	Visual
208	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Firing pin release.- Each firing pin shall release from the cocked position in accordance with the provisions of 3.2. Failure of any firing pin assembly to comply with 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- Each initiator pin shall be tested for security with the safety pin in each position. Failure of any initiator pin to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- The chamber and cap subassembly shall withstand a breakaway torque application as provided in 3.4. Failure of any subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.2.3.4 Residual Magnetism.- Each initiator assembly shall be subjected to the residual magnetism check. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.2.3.5 Functional.- The following test shall be performed by the Government.

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4.2.3.5.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size	AQL
-65° to 70°F	7	0.65%
65° to 75°F	7	0.65%
200° to 205°F	7	0.65%

4.2.4 Inspection Equipment.- The examination on tests shall be made using equipment listed on IEL 10551995.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (cap end south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

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4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within five minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.5.1.1.

4.3.5.1.1 Temperature conditioning.- A standard XM92 initiator loaded with a dummy M155 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. Preparation for Delivery

5.1 Packing and marking.-Packing and marking shall be in accordance with drawings C8798142 and C8798143.

6. Notes

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

Custodian:

Army - MU

Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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components and materials used were manufactured, tested, and inspected in accordance with the

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